

Meezaan Chishty

AI/ML Engineer | Model Training & Applied AI

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PROFESSIONAL SUMMARY

AI/ML Engineer with hands-on experience building production systems, fine-tuning domain-specific LLMs, and deploying applied AI pipelines. Shipped enterprise platforms combining multi-agent RAG workflows, visual document extraction, and FastAPI microservices. Proven track record of cross-functional and asynchronous remote collaboration across international teams in the US, China, and India.

EXPERIENCE

Full Stack AI/ML Engineer — FBSPL

Feb 2025 – Present

- Leading development of a domain-adapted LLM for insurance policy underwriting and document intelligence, designing the fine-tuning workflow, dataset preparation pipeline, and evaluation benchmarks while maintaining strict proprietary data boundaries.
- Architected and shipped **FBSPLAI** (fbsplai.com) — a production AI insurance sales platform with **3 Next.js 14 portals** and **10 FastAPI microservices**, orchestrating multi-agent RAG workflows for live client quote comparisons.
- Engineered an ACORD document extraction pipeline combining computer vision, visual document models, and OCR; automated extraction of tabular data, checkboxes, and form fields into structured databases.
- Delivered a distributed real-time operations monitoring platform across **700+ endpoints** with sub-**200 ms** WebSocket broadcast updates, providing operations teams with live endpoint health and status visibility.
- Introduced asynchronous task queues (Celery + Redis) and containerized microservices with Docker, improving background task processing throughput and deployment reliability.

Python Developer — NeoPanda Tech

Mar 2024 – Feb 2025

- Built and maintained production RESTful APIs and backend microservices in Python, integrating automated AI pipelines into daily content workflows.
- Built a RAG-based news intelligence system using vector embeddings and an autonomous AI agent; achieved a **99.9% deduplication rate** across incoming articles using MinHash and cosine similarity, reducing manual newsletter curation time by **85%** (from 4 hours daily to under 35 minutes).
- Developed a robust web crawler and scraper suite with proxy pooling, automated retry logic, and Playwright browser fallback, ingesting **5,000+ daily articles** from **50+ regional sources** into a SQLite pipeline.

AI/ML Engineering Intern — IIT Roorkee

Jan 2024 – Feb 2024

- Trained and evaluated supervised NLP classification models for sentiment analysis and emotion detection using Scikit-Learn and Pandas, achieving **>85% classification accuracy** on validation sets.
- Built regression models for small-business sales forecasting, evaluating predictions using Root Mean Squared Error (RMSE) to capture seasonal demand patterns.
- Benchmarked Small Language Models (SLMs) against larger LLMs on parameter efficiency and inference latency to guide model selection for resource-constrained edge deployments.

Lead Product Engineer — Zume, Inc.

Apr 2023 – Mar 2024

- Led AI application and ML feature development across an Agile/Scrum workflow, owning end-to-end delivery for **3+ core product features** with cross-functional US stakeholders.
- Engineered **Vocalytics-V0.3** — a voice-driven BI tool converting spoken natural-language queries into interactive Plotly BI visual workflows via speech recognition and OpenAI GPT-4, coupled with LightGBM sales forecasting.
- Implemented AST-based syntax validation to restrict executable operations before runtime evaluation in dynamic analytics workflows; packaged the desktop application using cx_Freeze.

TECHNICAL SKILLS

Model Training & Evaluation: PyTorch, Hugging Face Transformers, Supervised Fine-Tuning (SFT), Model Evaluation (Accuracy, F1-Score, RMSE), Scikit-Learn, LightGBM, SLM Benchmarking.

Applied AI & LLM Systems: Retrieval-Augmented Generation (RAG), Multi-Agent Workflows (LangChain, LangGraph), Prompt Engineering, Semantic Search, Computer Vision, OCR (Cloud Vision, Tesseract), OpenAI & Gemini APIs, Local Inference (Ollama).

Software Engineering: Python, TypeScript, JavaScript, SQL, Go, Bash, FastAPI, Next.js 14, React.js, Flask, Node.js, Express.js, REST APIs, WebSockets, Celery, Pydantic, Turborepo, Radix UI.

Data & Infrastructure: Vector Databases (Qdrant, pgvector, Weaviate), Kùzu Graph Database, PostgreSQL, MongoDB, Redis, SQLite, Docker, AWS (EC2, S3, ECS), Nginx, Git, Linux, CI/CD.

SELECTED PROJECTS

FBSPLAI — Enterprise AI Insurance Platform — *Proprietary / Client Live* 2025

<https://fbsplai.com>

- Orchestrated 10 FastAPI microservices integrating multi-provider LLMs (OpenAI, Google Gemini) and Cloud Vision OCR for document intelligence and conversational insurance quote advisory.
- Built with Celery task queues, Redis caching, and MongoDB on AWS EC2 behind an Nginx reverse proxy with JWT/RBAC security, powering 3 Next.js 14 client and underwriter portals.

Hippo Cortex Retrieve — Cognitive Long-Term AI Memory Engine — *Open-Source Implementation* 2026

<https://github.com/Mizan-muf/Hippo-Cortex-Retrieve>

- Working open-source architecture that retains and retrieves context across long-running AI agent conversations using a dual-retrieval pipeline.
- Combines **Qdrant** for semantic dense vector search, **Kùzu Graph Database** for entity-relation traversal, and an immutable Markdown file store for transparent auditability, integrated with local LLMs via Ollama.

Vocalytics-V0.3 — Voice-Driven Sales Intelligence Platform — *Open-Source Analytics Tool* 2024

<https://github.com/Mizan-muf/Vocalytics-V0.3>

- Voice-to-BI analytics workflow converting spoken natural-language commands into structured data queries, automated Plotly visual charts, and narrative summaries using OpenAI GPT-4.
- Trained sales forecasting models with LightGBM and built AST-validated code parsing to restrict dynamic operations, packaged for desktop deployment via cx_Freeze.

Additional projects: mizanchishty.ddns.net/#projects • github.com/Mizan-muf

EDUCATION

Jamia Hamdard University Jan 2025 – Present
Master of Computer Applications (MCA) in Computer Science **GPA: 8.95 / 10**

Manipal University Jaipur Oct 2021 – Apr 2024
Bachelor of Computer Applications (BCA) **GPA: 8.88 / 10**
Dean's List Awardee — Highest GPA in cohort

SELECTED CERTIFICATIONS

- **Deep Learning: Model Optimization & Tuning** — LinkedIn Learning ([Credential](#))
- **Deep Learning with TensorFlow** — LinkedIn Learning ([Credential](#))
- **Hands-On AI: Building LLM-Powered Apps** — LinkedIn Learning ([Credential](#))
- **Google Cloud Platform for Machine Learning** — LinkedIn Learning ([Credential](#))
- **Career Essentials in Generative AI** — Microsoft & LinkedIn ([Credential](#))
- **Core Java Specialization** — LearnQuest / Coursera ([Credential](#))